

Home Is Where the Points Are

Arsenal FC Home vs. Away Performance · Premier League 2015-16 to 2024-25

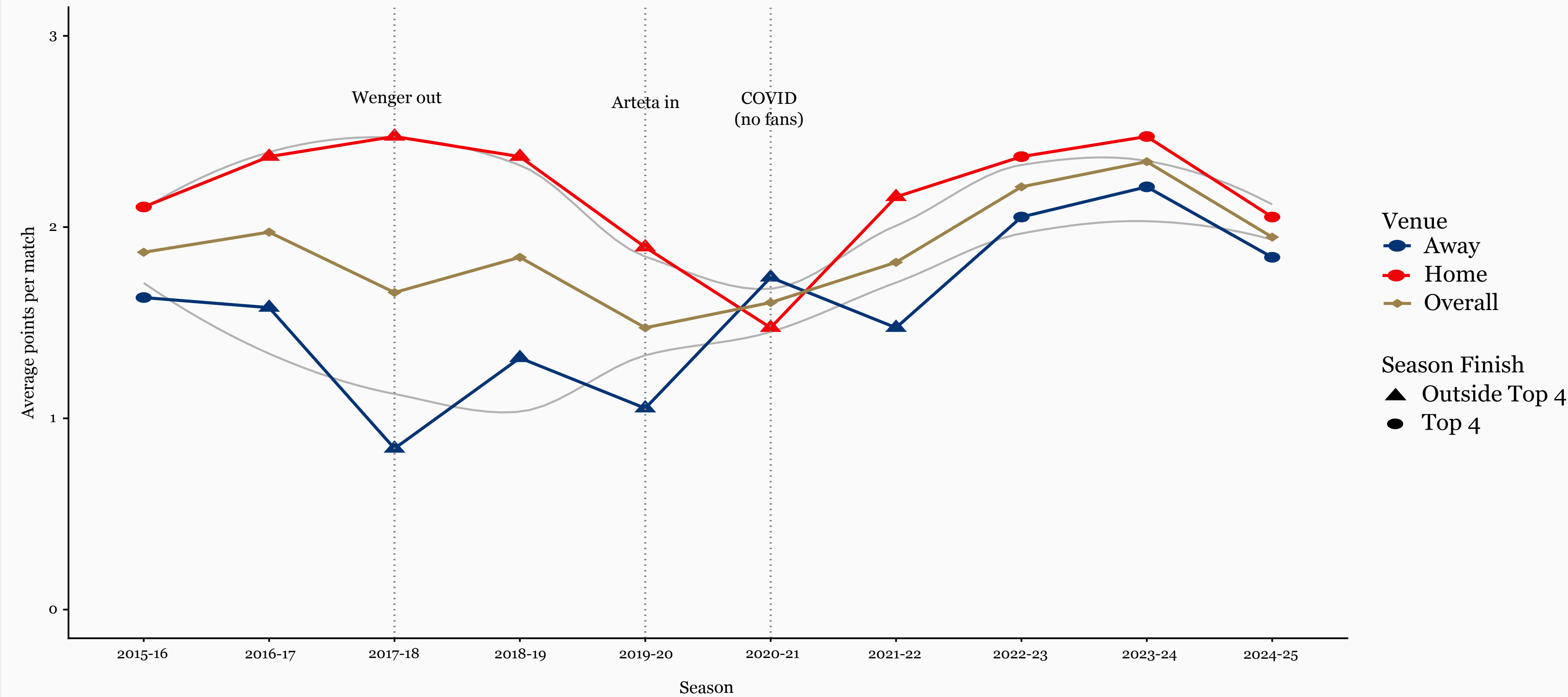
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IST 421 M002 Spring 2026
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DATA
380 Arsenal matches across 10 Premier League seasons (2015–16 to 2024–25), sourced from football-data.co.uk. Variables include goals, results, venue, and date. Filtered from ~3,800 matches to 380 rows and 11 columns.

QUESTION
Home advantage is well-documented in football, but its structural causes remain debated. Does Arsenal's home advantage represent a statistically significant difference in points per match — and does crowd presence explain it?

STORY & MOTIVATION
Arsenal's home advantage is statistically real and persistent. The COVID-19 season, played without fans, collapsed it league-wide — isolating crowd effect as the structural driver. Football analysts, club strategists, and sports journalists use these insights to inform scheduling, fan policy, and performance planning.

POINTS PER MATCH BY SEASON: HOME VS. AWAY VS. OVERALL



HYPOTHESIS AND RESULTS

H_0 : No difference in home vs. away performance

H_1 : Performance differs by venue

A two-sample t-test compared Arsenal's average points per match at home vs. away.

With $p = 0.0000057$, we reject the null hypothesis (H_0) Arsenal earn 0.34 to 0.86 more points per match at home — a statistically significant and persistent advantage.

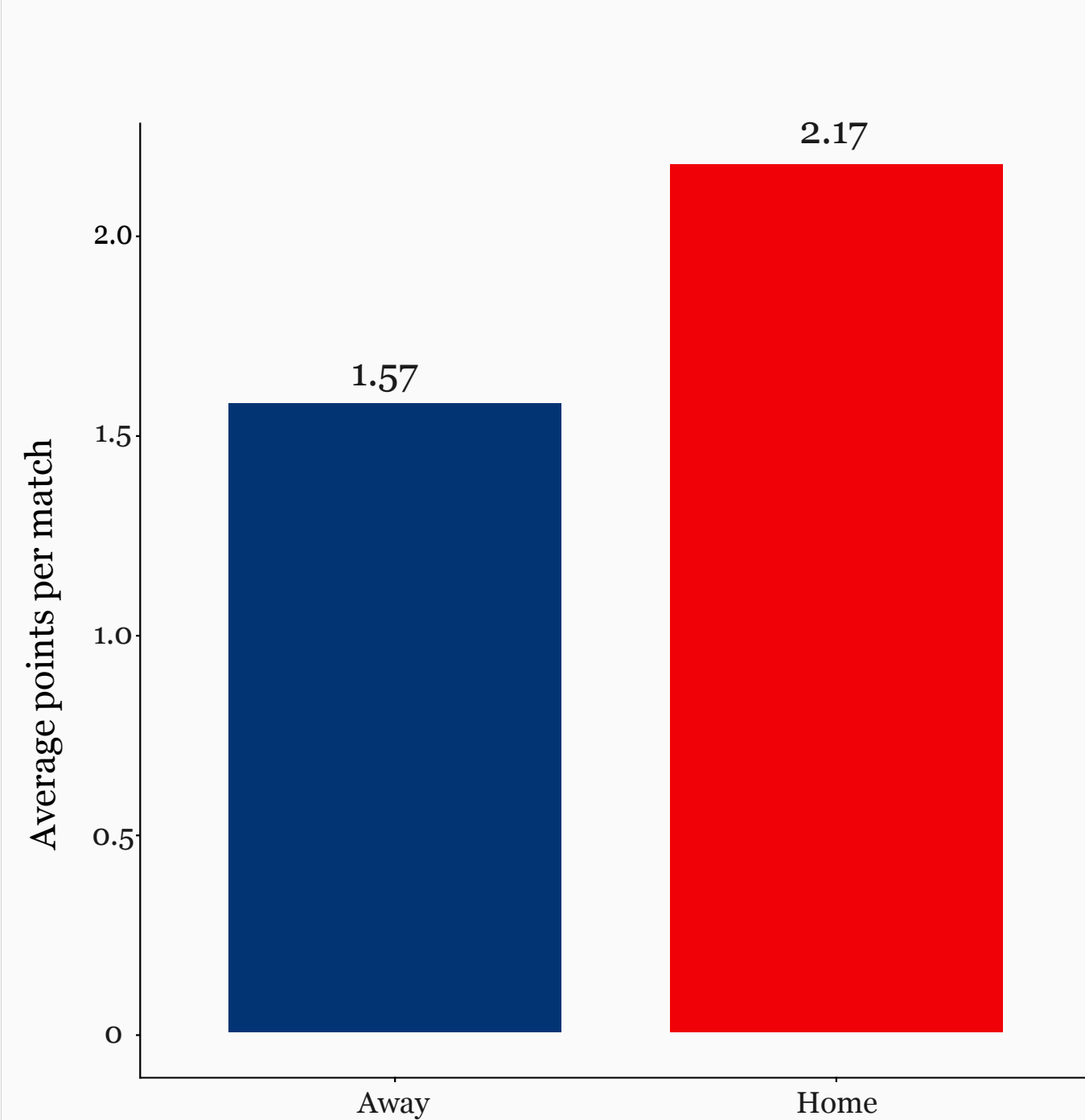
ABOUT THIS POSTER

The line chart tracks Arsenal's points per match across 10 seasons, annotated at three turning points: Wenger's exit, Arteta's arrival, and the COVID no-fans season.

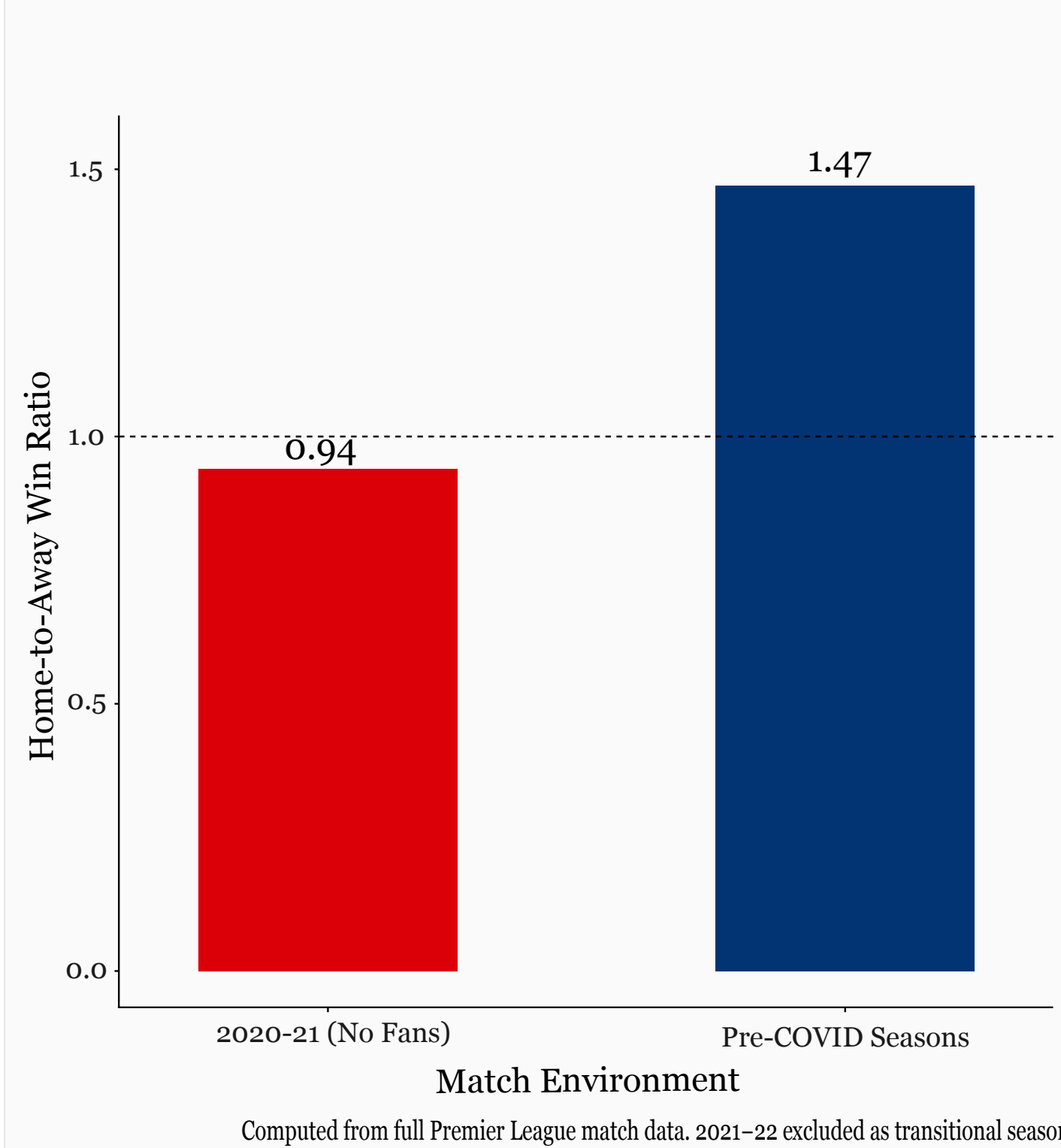
The bar charts below tell the supporting story — how much the points gap is, whether it shows up in goals scored or conceded, and whether it held league-wide or was unique to Arsenal.

The COVID season is the natural experiment: when fans were removed, home advantage collapsed across the entire Premier League — confirming that crowd effect is structural and not coincidental.

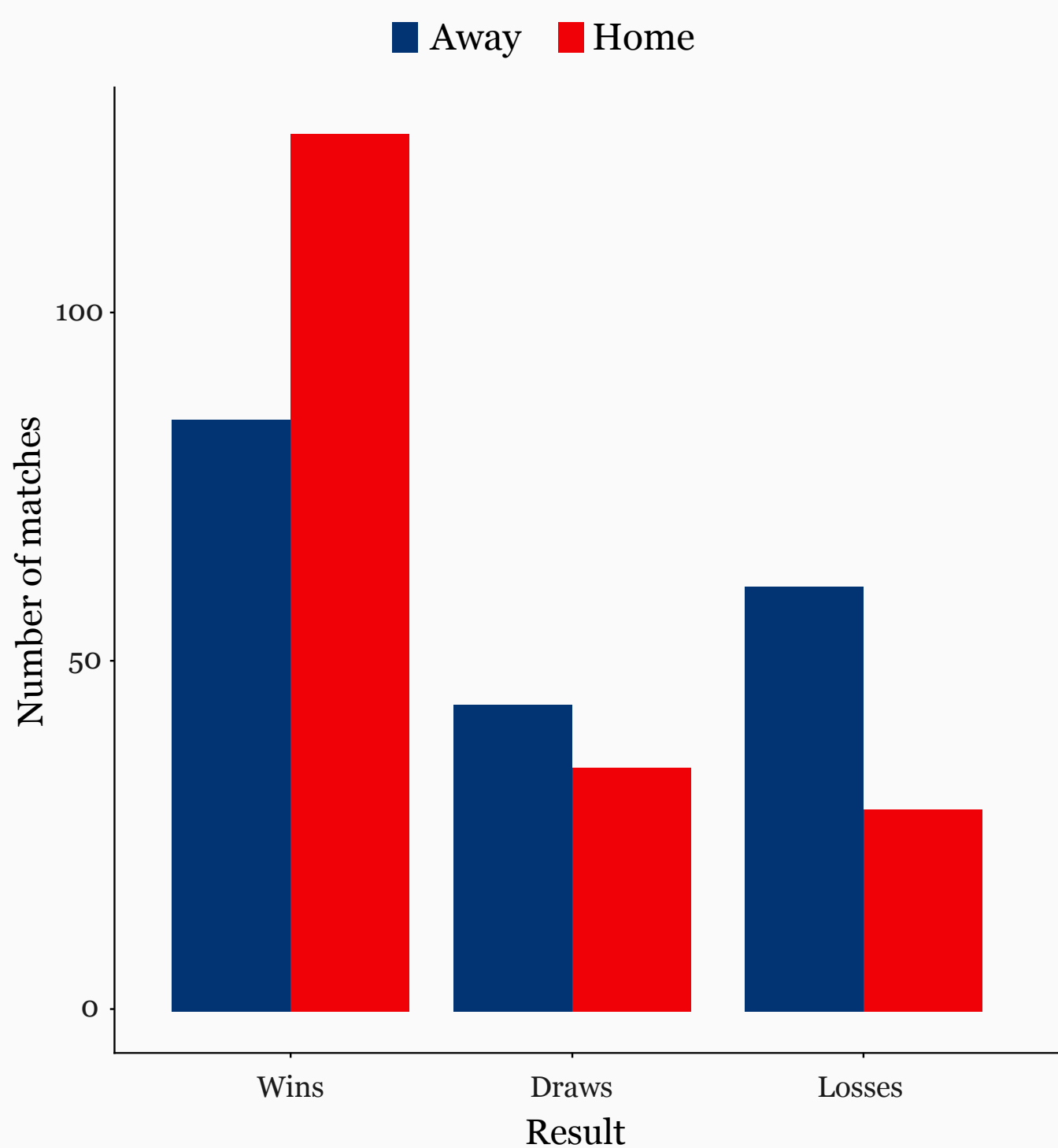
AVERAGE POINTS PER MATCH



LEAGUE-WIDE HOME ADVANTAGE



MATCH OUTCOMES BY VENUE



GOALS SCORED AND CONCEDED

